

$$H_{TP}(\omega) \quad \bullet \text{---} \circ \quad h_{TP}(t)$$

$$T_s rect_{\omega_c}(\omega) \quad \bullet \text{---} \circ \quad \frac{\sin(\omega_c t)}{\omega_c t}$$

Eckfrequenz: $f_c = \frac{\omega_c}{2\pi} = \frac{1}{2T_s}$

Verstärkungsfaktor: $k = T_s$

$$m(t) = m_s(t) * h_{TP}(t)$$

$$= \left(\sum_{n=-\infty}^{\infty} m(nT_s) \cdot \delta(t - nT_s) \right) * \frac{\sin(\omega_c t)}{\omega_c t}$$

$$= \sum_{n=-\infty}^{\infty} m(nT_s) \cdot \frac{\sin(\omega_c(t - nT_s))}{\omega_c(t - nT_s)}$$